

Nonlinear Add/Remove Drift, Interaction Count, and a Possible Bridge to G

Matter-as-machine

Abstract

This paper describes a discrete add/remove mechanism in which a small nonlinear asymmetry between addition and removal may produce a systematic drift of the ensemble center when the excursion asymmetry is directionally selected and persistently stored as a velocity-state variable. The dimensionless drift state is hypothesized to grow with the number of effective interactions. Once a physical velocity scale and interaction rate are specified, this growth becomes physical acceleration. The paper then states the minimal conditional bridge required to connect this acceleration to Newtonian gravity and the gravitational constant G.

The central result is not that gravity has been derived. The central result is a conditional kinematic bridge:

if $u(n)$ is proportional to n/N^2
and $n(t)$ is produced locally at rate $\lambda(r)$,
and physical velocity is $V(t) = V_* u(n(t))$,
then $a = dv/dt$ is proportional to $V_* \lambda(r)/N^2$.

To match Newtonian gravity, the local effective interaction rate must have inverse-square form:

$$\lambda(r) = \lambda_0 M / r^2.$$

Then:

$$a \sim V_* C \lambda_0 M / (N^2 r^2).$$

Comparing this with $a = GM/r^2$ gives:

$$G \sim V_* C \lambda_0 / N^2.$$

Thus, in this interpretation, G would measure the normalized production rate of effective interactions per source mass, after the physical velocity scale V_*

and internal N^2 suppression are accounted for. The r^2 dependence belongs to $\lambda(r)$, not to G. The inverse-square law is therefore not derived by the

add/remove asymmetry alone; it must come from a separate three-dimensional flux-density or vector-rate rule.

0. Motivating analogy: frogs and floating islands

A simple thought experiment motivates the add/remove mechanism.

Imagine frogs of mass m jumping over a floating island of mass M . Each jump transfers the same impulse p .

When a frog lands, the island is temporarily heavier:

$$M + m.$$

The same impulse then produces a smaller velocity change:

$$\Delta v_{\text{land}} = p/(M+m).$$

When a frog takes off, the island is temporarily lighter:

$$M - m.$$

The same impulse then produces a larger velocity change:

$$\Delta v_{\text{takeoff}} = p/(M-m).$$

Because:

$$p/(M-m) > p/(M+m),$$

the takeoff and landing effects do not cancel exactly. Over many jumps, a small second-order drift can appear opposite to the average frog motion.

Now consider two islands. Each island emits its own momentum carriers. Carriers move both ways between the islands. If the timing and direction of exchange are not perfectly symmetric, the same inverse-mass response can create a residual mutual drift.

This analogy is not the proof of the model. It only illustrates the physical

intuition behind the formal add/remove rule used below: adding quanta increases

the effective denominator, removing quanta decreases it, and equal-looking exchange events can therefore have unequal velocity effects.

1. Definitions

Consider an ensemble of identical target particles. Each particle is represented by a finite internal count N . The count N is interpreted as an effective number of internal instructions, quanta, or discrete degrees of freedom participating in the response of the particle.

Let:

N = base internal count of the target particle
 k = temporary imbalance around N
 n = number of effective interactions experienced by the particle
 t = time
 λ = $\lambda(r) = dn/dt$, the local effective interaction rate [s^{-1}]
 λ_0 = distance-independent interaction-production coefficient per unit source mass [$m^2 \text{ kg}^{-1} s^{-1}$], defined by $\lambda(r) = \lambda_0 M/r^2$
 $u(n)$ = dimensionless drift state as a function of interaction count
 V_* = physical velocity scale converting u into physical velocity [$m \text{ s}^{-1}$]
 $V(t)$ = physical velocity of the ensemble center, $V(t) = V_* u(n(t))$
 $X(t)$ = physical position of the ensemble center
 M = source mass
 r = distance from the source
 G = gravitational constant

The analysis is written for an ensemble center. The same reasoning can be applied to the median position if the distribution remains sufficiently regular that the median tracks the same systematic drift as the mean. In finite samples, the mean is usually smoother, while the median is more robust but noisier.

2. Exchange invariant

The interaction picture assumed here is an add/remove exchange picture:

interaction changes particle size/energy by adding or removing one quantum.

An event is not a silent replacement of one internal quantum by another. The basic events are:

add: the particle gains one quantum, so its effective count increases.
remove: the particle loses one quantum, so its effective count decreases.

Replacement-like behavior, if needed, must be represented as two exchange events:

remove A, then add B.

This matters because the mechanism depends on the different response of a particle with effective count $N-k$ and a particle with effective count $N+k$.

3. Local nonlinear asymmetry

Suppose a unit exchange event produces a response proportional to the inverse of the effective count. Then the same unit event has different magnitude depending on whether the particle is temporarily smaller or larger.

Compare two nearby effective counts:

$N - k$
 $N + k$

The difference between the two inverse responses is:

$$\begin{aligned} & 1/(N-k) - 1/(N+k) \\ &= ((N+k) - (N-k)) / ((N-k)(N+k)) \\ &= 2k / (N^2 - k^2). \end{aligned}$$

For $|k|$ much smaller than N :

$$1/(N-k) - 1/(N+k) \sim 2k / N^2.$$

Therefore the add/remove asymmetry is suppressed by N^2 .

This is the mathematical core of the mechanism. It does not require any gravitational interpretation.

4. Excursion scaling and the missing stochastic expectation

The local asymmetry derived above can be accumulated over a coherent excursion. Consider two opposite excursions of equal length m .

In a removal-dominated excursion, each subsequent removal acts on a smaller effective count:

$$1/(N-1), 1/(N-2), 1/(N-3), \dots, 1/(N-m).$$

In an addition-dominated excursion, each subsequent addition acts on a larger effective count:

$$1/(N+1), 1/(N+2), 1/(N+3), \dots, 1/(N+m).$$

The cumulative difference between the two excursions is:

$$D_m = \sum_{j=1..m} [1/(N-j) - 1/(N+j)].$$

Each term is:

$$\begin{aligned} & 1/(N-j) - 1/(N+j) \\ &= 2j / (N^2 - j^2). \end{aligned}$$

For m much smaller than N:

$$1/(N-j) - 1/(N+j) \sim 2j / N^2.$$

Therefore:

$$\begin{aligned} D_m &\sim \sum_{j=1..m} 2j/N^2 \\ &= (2/N^2) \sum_{j=1..m} j \\ &= (2/N^2) m(m+1)/2 \\ &= m(m+1)/N^2. \end{aligned}$$

For $1 \ll m \ll N$:

$$D_m \sim m^2/N^2.$$

Thus a coherent excursion of length m produces a nonlinear excess of order:

$$m^2/N^2.$$

If the effective excursion scale of a random walk after n interactions is:

$$m \sim \sqrt{n},$$

then:

$$\begin{aligned} D_m &\sim (\sqrt{n})^2 / N^2 \\ &\sim n / N^2. \end{aligned}$$

This suggests the effective dimensionless drift scaling:

$$u(n) \sim C n / N^2,$$

where C absorbs the efficiency with which random-walk excursions are selected, oriented, and converted into directed center drift.

This derivation is conditional. It requires that the excursion asymmetry is directionally selected and survives ensemble averaging. If excursions occur with exactly symmetric signs, the drift cancels. If the response saturates, or if the imbalance diffuses into changes of N rather than center motion, the scaling will change.

Therefore this section should be read as a scaling ansatz derived from coherent excursions, not yet as a full stochastic expectation theorem. A complete proof

must specify a stochastic rule for the sequence of add/remove events and compute:

$$E[u_n].$$

Such a proof must account for excursion-length distributions, returns, first-return structure, local times, correlations, sign selection, and cancellation between oppositely oriented excursions. In that complete calculation, C should not hide arbitrary dynamics; it should be computable from the stochastic rule, or at least bounded.

This is the main mathematical burden of the model. It is not a secondary correction to the bridge. Until $E[u_n]$ is computed from a concrete stochastic

process, the law $u(n) \sim C n/N^2$ should be treated as the target scaling suggested by the excursion calculation, not as a theorem.

The required target result is:

$$E[u_n] \sim C n/N^2$$

in the weak-excursion regime. If the expectation instead scales as $\sqrt{n}$, saturates, or vanishes, the proposed bridge to constant acceleration changes or fails.

The quantity $u(n)$ is not yet a physical velocity. It is an interaction-indexed dimensionless drift state. A physical velocity scale V_* is required:

$$V(t) = V_* u(n(t)).$$

If one defines an interaction-indexed displacement $q(n)$ by summing u over interaction steps, then:

$$\begin{aligned} q(n) &\sim C (1 + 2 + \dots + n) / N^2 \\ &\sim C n(n+1) / (2N^2). \end{aligned}$$

This $q(n)$ is not the physical position unless one also specifies the physical time or length per interaction step. The physical position is instead:

$$X(t) = \int V(t) dt.$$

The important point for the gravity bridge is that larger N suppresses the dimensionless drift state $u(n)$, and therefore the physical velocity $V(t)$, as

$$1/N^2 \text{ for the same number of effective interactions.}$$

5. From interaction count to acceleration

The variable n is an interaction count, not a time coordinate.

If effective interactions arrive at a local rate λ , with λ understood as $\lambda(r)$ held constant during the interval:

$$\lambda = dn/dt,$$

then:

$$n(t) = \lambda t$$

for constant λ .

Substituting this into the drift law:

$$u(t) \sim C \lambda t / N^2.$$

This step contains an important memory assumption. The variable u is treated as a persistent velocity-state variable, not merely as a per-excursion displacement bias. Equivalently, over the interval being modeled:

$$du/dn \sim C/N^2.$$

The bridge therefore assumes that this drift state is not immediately relaxed, decorrelated, reset, or absorbed into unrelated internal changes. A more general state equation would be:

$$du/dt \sim C \lambda(r)/N^2 - \Gamma u + \text{noise},$$

where Γ is a relaxation or reset rate. The acceleration bridge below is the weak-field, unsaturated, persistent-memory limit in which the drive term dominates over the interval of interest, or in which the steady-state response is handled separately.

Physical velocity is:

$$V(t) = V_- \cdot u(t)$$

so:

$$V(t) \sim V_- \cdot C \lambda t / N^2.$$

Acceleration is the time derivative of physical velocity:

$$a = dV/dt.$$

Thus:

$$a \sim V_- \cdot C \lambda / N^2.$$

This is the central bridge from interaction count to acceleration:

more effective interactions per second produce larger acceleration of the ensemble center or median center.

In this framework, acceleration is not introduced as a primitive force. It is the rate at which the interaction-driven physical velocity grows.

5b. Numerical evidence for the current stochastic rule

The current live simulation uses the following concrete stochastic rule:

```
if coin:
    k += 1
    v -= 1/(N-k)
else:
    k -= 1
    v += 1/(N-k)
x += v.
```

The visualization displays the attraction direction as $-x$ and $-v$. Therefore a positive displayed drift corresponds to negative raw x and v in the update rule.

For this specific rule, the sign of the mean velocity drift can be checked directly. Conditional on the previous internal state k , the two possible events give:

$$\begin{aligned} E[\Delta v \mid k] &= (1/2) [1/(N-k+1) - 1/(N-k-1)] \\ &= -1 / ((N-k)^2 - 1). \end{aligned}$$

Therefore the displayed velocity increment is:

$$E[\Delta v_{\text{display}} | k] = 1 / ((N-k)^2 - 1).$$

In the weak-excursion regime $k \ll N$:

$$E[\Delta v_{\text{display}} | k] \approx 1/N^2.$$

Thus:

$$E[v_{\text{display}}(n)] \approx n/N^2$$

and, because x accumulates velocity:

$$E[x_{\text{display}}(n)] \approx n(n+1)/(2N^2).$$

This is not yet a derivation of gravity. It is evidence that the current stochastic toy rule really does contain the second-order drift used by the bridge.

A C++ stress test is included at:

`src/nonlinear_random_walk_drift.cpp`

Code provenance:

`f15edbff493b83a9595506a30ef1910fa6628af7`

with a summary and generated outputs saved in:

`results/RESULTS.txt`
`results/nonlinear_random_walk_drift.txt`

Representative results:

```
N=1000, walkers=8000, ticks=5000, runs=30
  positive mean x: 30/30
  positive median x: 30/30
  average displayed mean v: 0.005139, theory 0.005000
  average displayed mean x: 12.7788, theory 12.5025

N=10000, walkers=8000, ticks=20000, runs=16
  positive mean x: 15/16
  positive median x: 12/16
  average displayed mean v: 0.0001939, theory 0.0002000
  average displayed mean x: 2.00394, theory 2.00010

N=100000, walkers=8000, ticks=20000, runs=16
  positive mean x: 11/16
  positive median x: 9/16
  average displayed mean v: 0.000001508, theory 0.000002000
  average displayed mean x: 0.07296, theory 0.020001

N=10000, walkers=20000, ticks=20000, runs=8
  positive mean x: 8/8
  positive median x: 7/8
  average displayed mean v: 0.0002025, theory 0.0002000
  average displayed mean x: 2.38452, theory 2.00010
```

For $N=100000$ with the same finite ensemble size, the predicted signal is much smaller and the sign of individual runs becomes noisy. This is expected: the signal falls as $1/N^2$ while finite-sample noise does not disappear unless the

ensemble size or run count is increased. The detailed standard errors are reported in results/RESULTS.txt.

6. Bridge to Newtonian gravity

Newtonian gravity gives acceleration:

$$a = G M / r^2.$$

The interaction-count mechanism gives:

$$a \sim V_* C \lambda(r) / N^2.$$

Here $\lambda(r)$ is the local effective interaction rate at distance r .

To reproduce Newtonian gravity, $\lambda(r)$ must contain inverse-square dilution:

$$\lambda(r) = \lambda_0 M / r^2,$$

where λ_0 is a distance-independent source coefficient per unit mass.

A possible route to this dependence is ordinary three-dimensional flux dilution. If effective carriers or interaction opportunities are emitted by a localized source with total production proportional to M , then the same total

production is spread over a spherical area proportional to $4 \pi r^2$. The local carrier density, and therefore the local event rate, can then scale as

$$M/(4 \pi r^2). \text{ The factor } 4 \pi \text{ can be absorbed into } \lambda_0.$$

This is compatible with an internal taxicab picture: the internal instructions may live along discrete axes, but the source-target separation still exists in three-dimensional space. The three axes provide possible directions of instruction response; the r^2 factor comes from dilution over the surrounding

two-dimensional surface area. This explains where r^2 may enter, but it still requires an explicit carrier or flux model. The inverse-square dependence is not

derived from the add/remove asymmetry by itself.

Substituting:

$$a \sim V_* C \lambda_0 M / (N^2 r^2).$$

Newtonian gravity requires:

$$a = G M / r^2.$$

Therefore:

$$G \sim V_* C \lambda_0 / N^2.$$

If $C = 1$:

$$G \sim V_* \lambda_0 / N^2.$$

Therefore, in this model, G would not be the physical velocity itself. G would be the coefficient that relates source mass to the distance-independent production strength of effective interactions, after accounting for internal

N^2 suppression and the velocity scale V_* . The distance dependence is carried by $\lambda(r)$, not by G .

The scalar rate $\lambda(r)$ gives only an event frequency. By itself it does not produce gravitational vector behavior. A gravitational model must introduce a vector interaction flux or field whose direction is selected by the source-target geometry.

For a single point source, the required vector form is:

$$\mathbf{a}_{\text{vec}} \sim - V_* C \lambda_0 M \hat{\mathbf{r}} / (N^2 r^2),$$

where $\hat{\mathbf{r}}$ points from the source to the target. The minus sign denotes

acceleration toward the source. A microscopic model must therefore explain not only the rate $\lambda(r)$, but also how the add/remove asymmetry is oriented along the source-target direction.

For many sources, the required superposition rule is:

$$\mathbf{a}_{\text{vec}} \sim - (V_* C \lambda_0 / N^2) \sum_i M_i \hat{\mathbf{r}}_i / r_i^2.$$

Equivalently, the model needs a vector interaction-rate field:

$$\Lambda_{\text{vec}}(\mathbf{x}) = - \lambda_0 \sum_i M_i \hat{\mathbf{r}}_i / r_i^2,$$

with:

$$\mathbf{a}_{\text{vec}} \sim (V_* C / N^2) \Lambda_{\text{vec}}(\mathbf{x}).$$

Without such a vector flux rule, $\lambda(r)$ is only a scalar event rate and cannot by itself reproduce direction, cancellation, extended-source behavior, or Newtonian superposition.

For a discrete event model this requirement is stronger than writing a smooth field after the fact. If several sources are present, the model must explain how source-labeled carriers, or their effective vector rates, combine before an event is selected. Otherwise a scalar Poisson rate plus a later directional choice may fail at cancellation points, inside shells, or near extended sources.

7. Interpretation of G

Under this bridge, G can be read as:

normalized effective-interaction production per unit source mass.

More explicitly:

$$\lambda(r) = \lambda_0 M / r^2$$

and:

$$G \sim V_* C \lambda_0 / N^2.$$

The terms mean:

$\lambda(r)$ = local effective interactions per second at distance r
 λ_0 = distance-independent production coefficient per unit source mass
 $[m^2 \text{ kg}^{-1} \text{ s}^{-1}]$
 V_* = physical velocity scale converting u into V
 $[m \text{ s}^{-1}]$
 M = source mass

r^2 = geometric inverse-square dilution
 N^2 = internal suppression of target response
 C = efficiency of the add/remove excursion asymmetry

The units are consistent:

$V_* \lambda_0$ has units $(\text{m/s})(\text{m}^2/(\text{kg s})) = \text{m}^3/(\text{kg s}^2)$,

which are the units of G .

However, this expression creates both a universality requirement and a possible phenomenological signal. Since laboratory and Solar-System tests strongly constrain deviations from universal free fall, the combination

$V_* C / N^2$

must be approximately universal for ordinary local matter, or any target-dependence must be below experimental bounds.

This condition is sharp. If N varies between particle species, materials, or composite bodies, then ordinary free fall remains universal only if the product

$V_* C$ varies approximately as N^2 in the same regime, or if the relevant

effective N is not actually composition-dependent for ordinary matter. Without one of these compensations, the model predicts composition-dependent acceleration.

There is also a simpler local route to approximate universality. A macroscopic body is not accelerated as one indivisible object in this picture. Gravity-like

response acts on its constituent particles separately. If ordinary local matter is built from constituent particles with the same effective internal count and the same response law, then each constituent receives the same acceleration. A larger body contains more constituents, but it also has proportionally more inertial content, so the body-level acceleration remains approximately the same.

In that regime, differences between macroscopic bodies are hidden because their constituent particles are effectively the same.

At the same time, the model naturally permits an effective gravitational coupling:

$G_{\text{eff}} = V_* C \lambda_0 / N^2$

to depend on internal count, interaction efficiency, source structure, or environment. This dependence is not allowed to appear freely in local equivalence-principle experiments, but galaxies are a possible discovery regime:

if the compensating local universality of $V_* C/N^2$ changes with galactic

environment, source organization, density, or long-range interaction history, the result could appear as a dark-matter-like departure from strict Newtonian gravity rather than as ordinary composition dependence.

So the role of G is to connect three things:

source mass,
 interaction production rate,
 and target acceleration.

Short form:

n creates dimensionless drift state u .
 dn/dt turns u -growth into physical acceleration through V_* .
 G converts source mass into dn/dt at distance r , after V_* and N are fixed.

8. Mean position and median position

For an ensemble of many particles, the systematic drift can be measured by the mean position:

$$X_{\text{mean}}(t).$$

If the distribution is not strongly skewed or fragmented, the median position:

$$X_{\text{median}}(t)$$

should follow the same direction of drift. The median may be preferred when outliers are large, but it may also fluctuate more strongly in finite samples.

The proposed acceleration can be defined as:

$$a_{\text{mean}} = d^2 X_{\text{mean}} / dt^2$$

or, when the median is stable enough:

$$a_{\text{median}} = d^2 X_{\text{median}} / dt^2.$$

Under the excursion-derived scaling, conditional on directional selection,

increasing the effective interaction rate $\lambda(r)$ increases these accelerations:

$$\begin{aligned} a_{\text{mean}} &\sim V_* C \lambda(r) / N^2 \\ a_{\text{median}} &\sim V_* C_{\text{median}} \lambda(r) / N^2 \end{aligned}$$

where C_{median} may differ from C in finite or skewed distributions.

9. Regime of validity

The excursion expansion assumes:

$$m \ll N.$$

Since the random-walk estimate uses:

$$m \sim \sqrt{n},$$

the weak-excursion regime also requires:

$$n \ll N^2.$$

The bridge should therefore be read as a weak-field, unsaturated, finite-count approximation. It cannot be assumed to hold for arbitrarily long times, arbitrarily large interaction counts, or relativistic velocities.

With constant $\lambda(r)$, the model gives:

$$V(t) \sim V_* C \lambda(r) t / N^2.$$

This grows without bound if extrapolated indefinitely. A physical completion must therefore include saturation, relativistic limiting behavior, or a change of regime before $V(t)$ exceeds the allowed physical speed scale.

10. Energy and exchange accounting

The model uses real add/remove exchange events. Therefore energy accounting is not optional.

A complete physical model must specify:

1. where added quanta come from; 2. where removed quanta go; 3. whether repeated exchange changes the target's rest mass or internal energy; 4. why ordinary gravitational acceleration does not produce unacceptable heating, drag, mass drift, or rapid source depletion; 5. what balance process maintains stable matter under continuous exchange; 6. what carries the opposite momentum or recoil.

The last point is essential. Target-side acceleration toward a source cannot be

only bookkeeping inside the target. For an isolated source-target system, the complete exchange must preserve total momentum and center-of-mass behavior. A full model must therefore identify whether the recoil is carried by emitted quanta, absorbed by the source, stored in an intermediate field, or balanced by a return channel.

Without such an accounting, the mechanism remains a kinematic drift model rather than a full physical theory.

11. Universality and possible dark-matter-like departures

The target-response factor:

$$V_* \propto C/N^2$$

is central. If it is constant for ordinary matter in local experiments, the model can reproduce approximate universality of free fall. If it varies with internal count, interaction efficiency, source structure, or environment, then the model predicts an effective coupling:

$$G_{\text{eff}} = V_* \propto C \lambda_0/N^2$$

that can differ between regimes.

The local constraint is therefore stronger than merely saying that N may vary. If N is particle-dependent or composition-dependent, then either $V_* \propto C$ must

track N^2 closely enough that $V_* \propto C/N^2$ remains universal for ordinary matter,

or the model predicts experimentally constrained composition-dependent acceleration. A galactic departure should therefore be tied to a change of regime, not to an unconstrained material dependence that would already have been visible locally.

This is not automatically a failure. It is a testable fork:

1. In laboratory and Solar-System regimes, G_{eff} must be nearly universal to satisfy equivalence-principle and precision-gravity constraints.

2. In galactic regimes, changes in effective internal response or interaction organization could appear as extra gravity, modified inertia, or dark-matter-like rotation and lensing phenomenology. The relevant variation should be environmental, collective, or source-structure dependent, so that local composition tests remain protected while galaxy-scale dynamics can depart from the Newtonian value inferred from visible mass.

The burden on the model is therefore sharper:

explain why G_{eff} is locally universal,
and identify when and why it departs from universality.

12. Required premises

The bridge to G depends on premises that must be shown independently.

Premise 1: Physical particles have an effective internal count N .

N must be given a physical meaning: internal instructions, quanta, cycles, or another discrete count.

Premise 2: Add/remove interactions occur physically.

The model requires real exchange events in which a particle gains or loses one quantum. Silent replacement is not enough for this mechanism.

Premise 3: The add/remove imbalance creates a nonzero directed drift.

The algebra proves local asymmetry, but the full dynamics must show that the asymmetry does not average to zero.

Premise 4: Random-walk excursions are directionally selected.

The excursion calculation gives $D_m \sim m^2/N^2$. With $m \sim \sqrt{n}$, this produces $u(n) \sim C n/N^2$. The remaining physical premise is that these excursions are not paired with exactly opposite excursions that cancel them. The underlying interaction rules must therefore produce a nonzero mean orientation of the excursion asymmetry.

Premise 4b: The stochastic expectation $E[u_n]$ is computed.

The coherent-excursion argument gives a scaling candidate, but a complete model must define the stochastic add/remove process and compute or bound $E[u_n]$. The calculation must include excursion-length distributions, returns, local times, first-return structure, correlations, and cancellation between oppositely oriented excursions. The coefficient C should be derived or bounded from this process rather than used as an unrestricted fitting parameter. This is the central missing theorem of the mechanism.

Premise 5: The excursion scale $m \sim \sqrt{n}$ is valid in the relevant regime.

This is the usual random-walk scaling, but the model must show when it applies and when it is modified by saturation, correlations, or finite- N effects.

Premise 6: Effective interactions are produced at a rate λ .

The model needs a physical source for $\lambda = dn/dt$.

Premise 6b: A physical velocity scale V_* is defined.

The excursion calculation produces a dimensionless drift state $u(n)$. To obtain physical acceleration, the model must define the velocity scale V_* converting u into physical velocity $V(t) = V_* u(n(t))$.

Premise 6c: The drift state has persistent memory over the interval considered.

The acceleration bridge assumes $du/dn \sim C/N^2$ and treats u as a persistent

velocity-state variable. If relaxation, reset, or decorrelation is present, the model must include a term such as $-\Gamma u$ and show the regime in which the persistent-memory approximation is valid.

Premise 7: λ scales with source mass.

To match Newtonian gravity:

λ proportional to M .

Premise 8: λ has inverse-square distance dependence.

To match Newtonian gravity:

λ proportional to $1/r^2$.

This may arise from three-dimensional geometric dilution of carriers or interaction opportunities over area $4\pi r^2$, but it must be explicitly derived from a source and propagation model.

Premise 8b: The scalar rate is promoted to a vector flux.

A scalar event rate $\lambda(r)$ gives frequency but not direction, cancellation, or superposition. The model must define a vector interaction-rate field $\Lambda_{\text{vec}}(x)$ whose contributions from multiple sources add approximately as $\sum_i -M_i \hat{r}_i/r_i^2$. In a discrete event model, it must also explain how source-labeled carriers or vector rates are combined before event selection, so that cancellation points, shells, and extended-source fields are reproduced.

Premise 9: Local universality and nonlocal departures are separated.

Since the acceleration contains the target-response factor $V_* C/N^2$, the model must define the regime in which this combination is approximately universal and the regime in which its dependence on internal count, interaction efficiency, source structure, or environment becomes observable. If N is composition-dependent, local universality requires either $V_* C$ to compensate approximately as N^2 or the effective N to be universal for ordinary matter. Local deviations must satisfy equivalence-principle bounds. Galactic deviations may become a dark-matter-like signal only if they arise from a genuine change of regime, such as environment, collective organization, or long-range source structure.

Premise 10: Composite bodies inherit particle-level acceleration.

A body is made of many particles. The model must show that gravity-like response acts on the constituent particles separately and that body-level free fall is the aggregate of those particle-level responses. If ordinary local constituent particles have the same effective mass/count and response law, then different macroscopic bodies can fall alike even if the mechanism is not fundamentally composition-blind at every possible scale.

Premise 11: Photon behavior is compatible.

Photons of different energies appear to follow the same gravitational paths. The model must explain why the energy dependence cancels, is suppressed, or is otherwise unobservable for photons.

A candidate explanation is that photons are a special taxicab-symmetric case. If photons of different energies share the same direction proportions in taxicab instruction space, then increasing photon energy increases the number of participating quanta without changing the normalized directional response. In that case, the photon can have different energy while retaining the same effective bending direction. This symmetry claim must be derived from the photon-formation rules, not merely assumed.

Premise 12: Energy exchange is balanced.

Add/remove events change particle size or energy. The model must explain why gravitational acceleration does not imply observable heating, mass drift, drag, or rapid source depletion.

Premise 12b: Momentum and recoil are balanced.

If the target accelerates toward the source, the opposite momentum must be carried somewhere. A complete exchange model must preserve total momentum for isolated systems.

Premise 13: The measured value of G can be recovered.

The model must eventually compute or calibrate:

$$\begin{aligned}\lambda(r) &= \lambda_0 M/r^2 \\ G &\sim V_* C \lambda_0/N^2\end{aligned}$$

and compare the result with measured gravity.

13. Conclusions

The mathematical part of the mechanism is the nonlinear add/remove asymmetry:

$$1/(N-k) - 1/(N+k) = 2k/(N^2-k^2) \sim 2k/N^2.$$

This gives a natural weak effect because the response is suppressed by N^2 .

If effective interactions accumulate into a persistent velocity-state variable so that:

$$u(n) \sim C n/N^2,$$

and if the effective interaction count grows in time locally as:

$$n(t) = \lambda(r) t,$$

then the ensemble center accelerates:

$$a \sim V_* C \lambda(r)/N^2.$$

To match Newtonian gravity:

$$a = GM/r^2,$$

the local interaction rate must have inverse-square form:

$$\lambda(r) = \lambda_0 M/r^2.$$

Then:

$$a \sim V_* C \lambda_0 M/(N^2 r^2).$$

So:

$$G \sim V_* C \lambda_0/N^2.$$

Thus the proposed interpretation of G is:

G measures the distance-independent source interaction coefficient after the velocity scale V_* and internal N^2 suppression are accounted for. The r^2 dependence belongs to the local interaction rate $\lambda(r)$, not to G itself.

The corresponding vector form would require:

$$\mathbf{a}_{\text{vec}} \sim - V_* C \lambda_0 M \hat{r} / (N^2 r^2).$$

This paper does not claim that gravity has been derived. It identifies a specific mathematical mechanism and states the bridge needed to connect it to G. For the bridge to become a physical theory of gravity, the model must still derive the stochastic expectation $E[u_n]$, the direction-selection rule, the

memory/relaxation law for u , the vector interaction-rate field with Newtonian superposition, the inverse-square interaction rate from a three-dimensional

source/flux model, the regime in which G_{eff} is locally universal, the regime in which departures from universality become observable, energy and momentum

balance under add/remove exchange, and the regime in which the unsaturated approximation is valid.